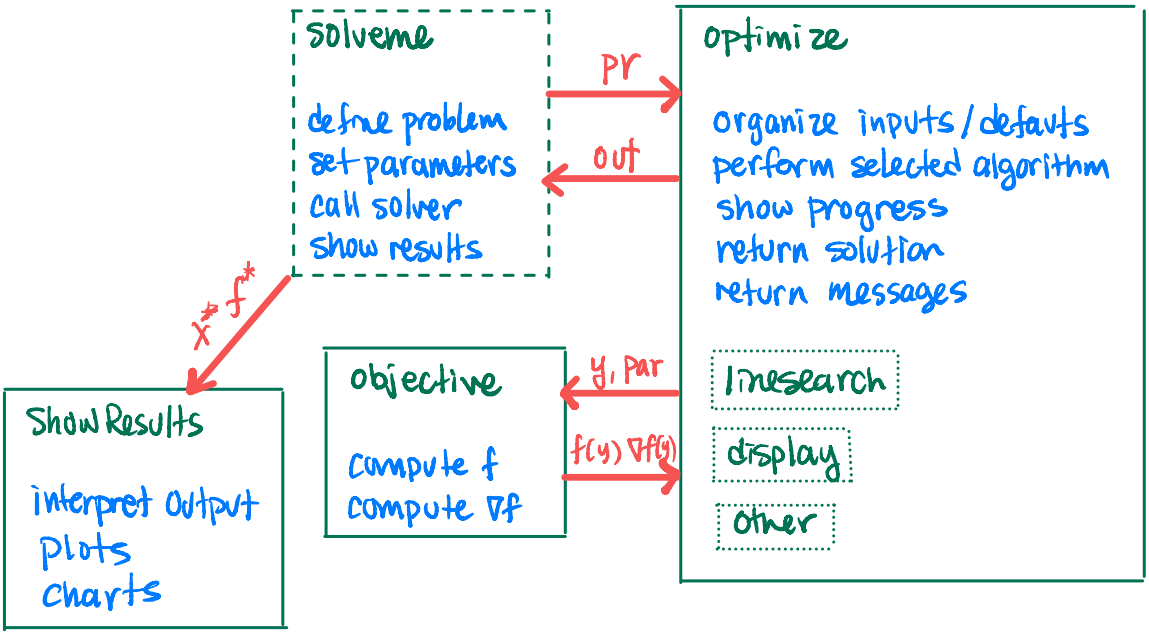
MATH 564 - September 3

VALVE DESIGN PROJECT

(15)	4.95	1.00	1.00	E = 7.8
(16)	2.50	3.00	3.30	E = 1.0
(17)	0	1	2.2	E = 27.0

In class we reviewed details of project #1. We currently have the background to complete items 1,2,3,4. Soon we will discuss how to implement the other items.

Code Organization Suggestion



pr	problem definition (matlab structure, python dictionary)
out	problem results (structure or dictionary)
par	parameter variable required by objective
y	test point in $\mathbb{R}^n$
$f(y)$	objective value at y
$\nabla f(y)$	gradient vector at y

Example Objective Function

```
function [f,g]=functionfit(x,par)

t=par(:,1);
v=par(:,2);

exarg=exp(-t/x(3));
sarg=(x(4)+x(5)*t).*t+x(6);
S=exarg.*sin(sarg);
vfit=x(1)+x(2)*S;
dv=vfit-v;
f=sum(dv.^2);

if nargout>1
    g=zeros(6,1);
    Sdv=dv.*S;
    Cdv=dv.*exarg.*cos(sarg);
    g(1)=sum(dv);
    g(2)=sum(Sdv);
    g(3)=(x(2)/x(3)^2)*sum(Sdv.*t);
    g(4)=x(2)*sum(Cdv.*t);
    g(5)=x(2)*sum(Cdv.*(t.^2));
    g(6)=x(2)*sum(Cdv);
    g=2*g;
end

return
```

Example Script file

```
clear pr

A=readmatrix('FFD.csv');
time=A(:,1);
voltage=A(:,2);

pr.obj=@functionfit;
pr.par=[time(:),voltage(:)];
pr.x0=[14;8;0.02;2000;1000;4]; % offset,amplitude,tau,omega,alpha,phi


pr.method='BFGS';
pr.linesearch='StrongWolfe';
pr.maxiter=10000;
pr.progress=5;
pr.ngtol=1E-10;
pr.dftol=1E-10;
pr.dxtol=1E-10;
pr.LS.c1=0.0001;
pr.LS.c2=0.9;
pr.CG.reset=0.2;

out=optu(pr);

ShowResults(out)
```

My "optimize" code documentation

```
function [out]=optu(pr)
% Function optu finds a local minimizer of an unconstrained function on R^n
% using a user-selected a gradient based method.
% Author: Tom Asaki
% Version: May 21, 2025
%
% Call:
%
%   [out]=optu(pr)
%
% Inputs:
%
%   pr  a structure variable containing all necessary problem information.
%       The various fields and [default values] are as follows.
%
%       .progress   A positive intgeger.  Progress will be displayed
%                   every (pr.progress) iterations.
%       .obj        function handle to the objective/gradient computation
%       .x0         vector of intial decision variable values.
%                   (or) n by p array of p initial points for NM or GA.
%       .par        variable to pass to objective function (for example
%                   containing parameters in a structure variable).
%       .method     string indicating optimization method.  Options:
%                       'GD'  (GradientDescent)
%                       'CG'  (ConjugateGradient)
%                       'BFGS' (quasi-Newton)
%                       'LBFGS' (Limited-Memory BFGS)
%                       'TR' (SR1 Trust Region and Steihaug-Toint steps)
%       .maxiter [inf]  maximum number of decision variable updates
%       .ngtol [1E-8]  stop tolerance on gradient norm
%       .dftol [1E-8]  stop tolerance on change in objective
%       .dxtol [1E-8]  stop tolerance on change in decision varable norm
%       .LS.method  string indicating the type of linesearch to perform
%                       'Armijo'       (appropriate for GD)
%                       'StrongWolfe'  (appropriate for CG and BFGS)
%       .LS.lambda [1]  line search initial step size multiplier
%       .LS.lambdamax [100] maximum line search step length
%       .LS.c1 [0.0001]  Armijo sufficient decrease parameter ( 0 < c1 < 1/2 )
%       .LS.c2 [0.9] Curvature condition parameter ( 0 < c1 < c2 < 1 )
%                 If using Conjugate Gradient method ( 0 < c1 < c2 < 1/2 )
%                 with default value [0.4]
%       .LBFGS.m [7] number of L-BFGS iterations to save
%       .CG.reset [0.1] orthogonality reset value for Conjugate Gradient
%       .TR.delta [1] initial trust region size
%       .TR.deltamax [100] maximum trust region size
%       .TR.deltatol [1E-8] stop tolerance on trust region size.
%       .TR.eta [0.01 0.25 0.75]  trust region parameters
%                  [ sufficient decrease , shrink , expand ]
%                  ( 0 <= eta1 < eta2 < eta3 < 1 )
%       .TR.maxcond [1000]  maximum condition number on approximate model
%                   hessian for trust region method.
%
% Outputs:
%
%   out     a structure variable containing all intial input values
%           and additional results of the optimization procedure.
%       .pr     copy of the input structure variable with added
%               default values and possibly some other algorithmic
%               necessary changes.
%       .x      (n by iter) array whose columns are the decision
%               variable iterates at each iteration
%       .f      (1 by iter) array whose entries are the corresponding
%               objective function values
%       .g      (n by iter) array whose columns are the gradient
%               vectors at each iteration.
%       .feval  total number of function evaluations
%       .geval  total number of gradient evaluations
%       .msg    output message - reason for algorithm termination
%
```